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SCHOOL OF COMPUTER APPLICATIONS

**IOT Project Report
on
Rain Detector Project**

Submitted in Partial fulfillment of the requirements for the award of the Degree of

MASTER OF COMPUTER APPLICATIONS

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1.Introduction to Project

A rain sensor clothes protection system is an automated solution designed to safeguard clothes drying outdoors from unexpected rainfall. Traditional drying methods rely on manual intervention, where individuals must quickly rush to collect clothes when rain begins. This approach is inconvenient, error-prone, and often results in clothes getting wet. To overcome these limitations, this project implements a smart clothes protection mechanism using a rain sensor, Arduino Uno, and a servo motor, providing a simple, low-cost, and efficient alternative for household use.

This project uses a **rain sensor module** to detect the presence of raindrops. When rainfall is detected, the sensor sends a signal to the **Arduino Uno**, which acts as the core controller. The Arduino then triggers a **servo motor** that automatically deploys a protective cover or mechanism to shield the clothes. Once the rain stops, the system retracts the cover, allowing clothes to continue drying naturally.

The system operates on a **binary detection mechanism**, where the rain sensor outputs a digital signal (LOW for rain, HIGH for no rain). This straightforward approach ensures reliable operation without requiring complex backend processing or network connectivity, making the system lightweight and practical for everyday use.

Overall, this rain sensor clothes protection project demonstrates how embedded systems and simple automation can be combined to solve real-life problems. By using basic hardware components such as an Arduino Uno, rain sensor, servo motor, and jumper wires, the project achieves automatic detection and protection in a cost-effective and scalable manner.

2.Components Used

Listed below are the components used in this project:

- Arduino Uno
- Rain Sensor Module
- Servo Motor (SG90 or equivalent)
- Jumper Wires
- Power Supply (USB or 5V Adapter)

3.Connectivity

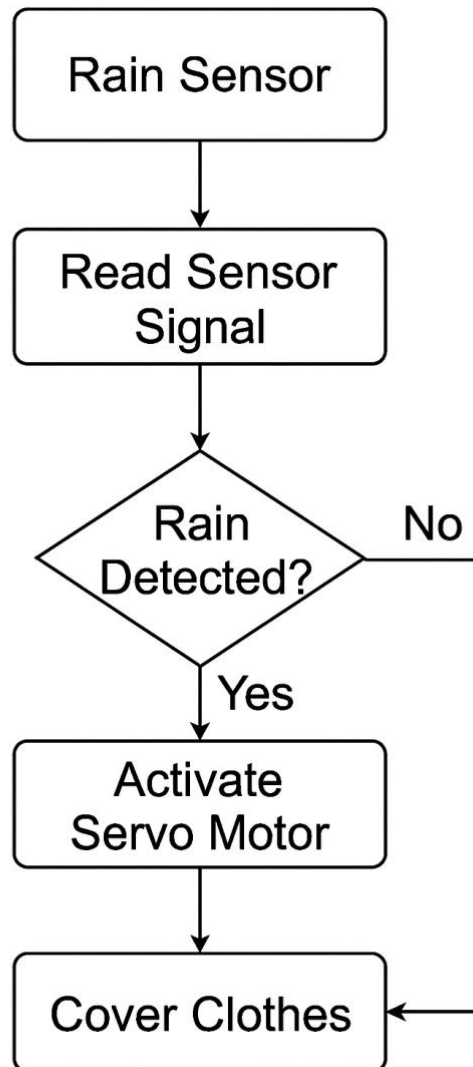
Arduino Uno to Rain Sensor:

Arduino Uno Pins	Rain Sensor Pins
D7	OUT
5V	VCC
GND	GND

Arduino Uno to Servo Motor:

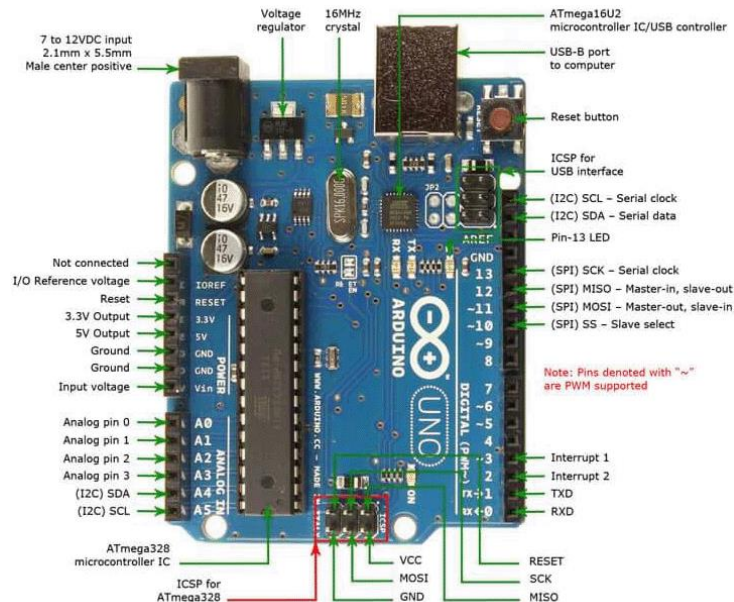
Arduino Uno	Servo Motor Pins
5V	VCC
GND	GND
D6	Signal

4.Dataflow



5.Description of Each Component

1. Arduino Uno



The Arduino Uno is an open-source microcontroller board based on the ATmega328P. It serves as the brain of the project, processing input signals from the rain sensor and controlling the servo motor accordingly. It features 14 digital I/O pins, 6 analog inputs, a USB connection, and a power jack, making it suitable for prototyping automation projects.

2. Rain Sensor Module



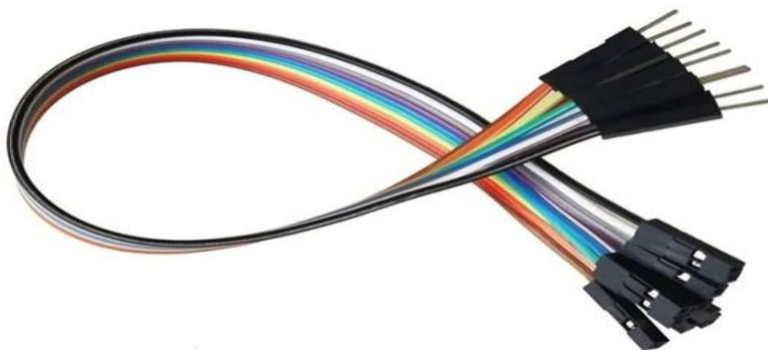
The rain sensor module detects the presence of water droplets. It consists of a sensing pad with conductive tracks and a control board. The sensor provides both analog and digital outputs, where the digital pin outputs **LOW** when rain is detected and **HIGH** when dry. This module is reliable for weather-based automation systems.

3. Servo Motor (SG90 or Equivalent)



The SG90 servo motor is a small, lightweight actuator that can rotate to a specific angle based on PWM signals from the Arduino. It has three connections: VCC, GND, and Signal. In this project, the servo rotates between 0° and 90° to deploy or retract the protective cover for clothes.

5. Jumper Wires



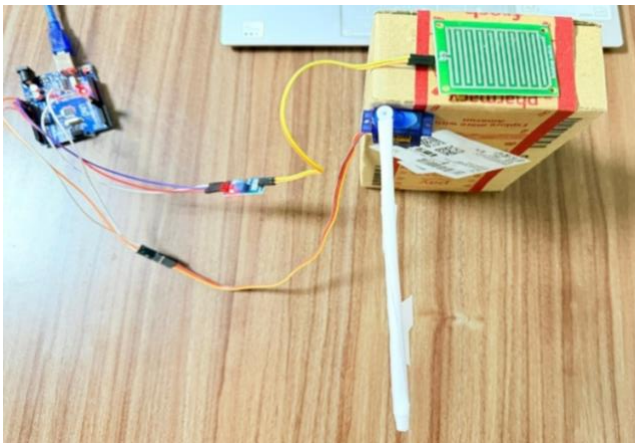
Jumper wires are used to establish electrical connections between the Arduino, rain sensor, and servo motor. They provide flexible and reusable wiring during prototyping, allowing easy modification and troubleshooting without soldering.

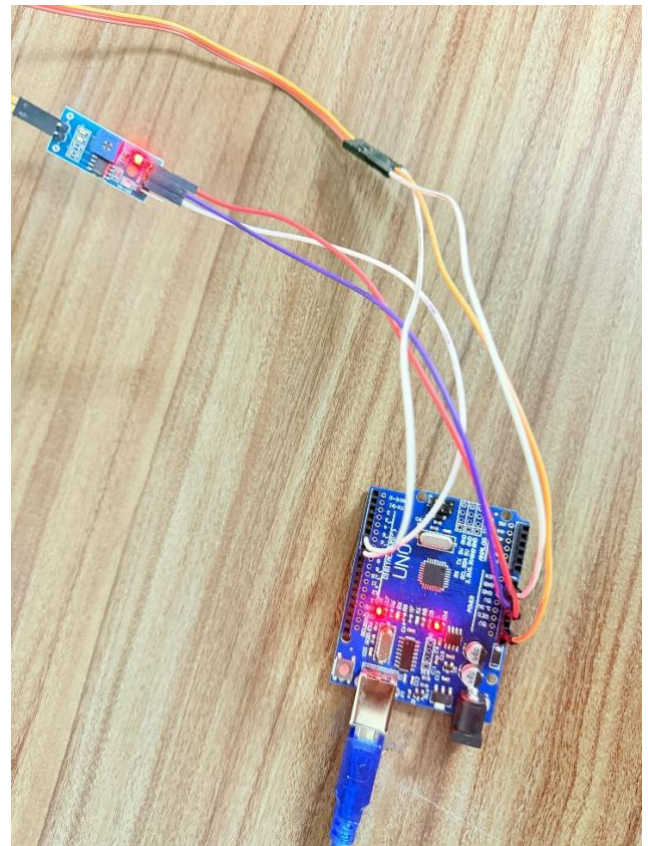
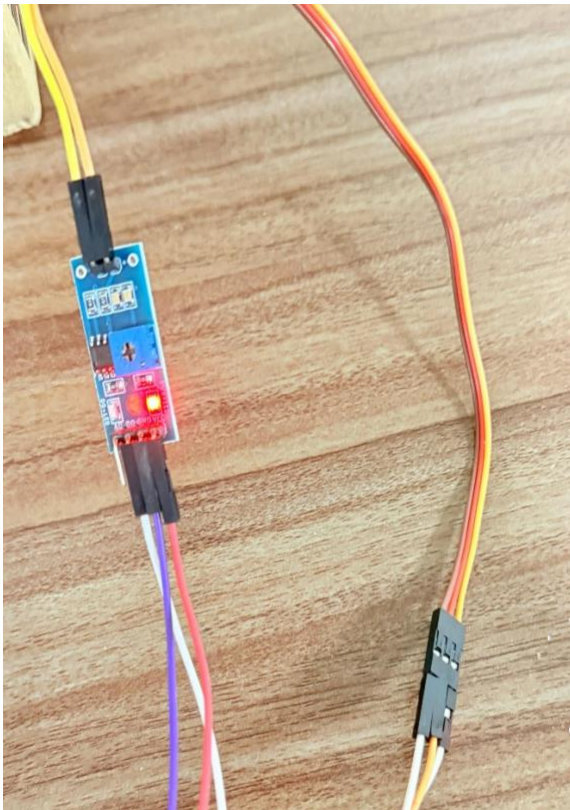
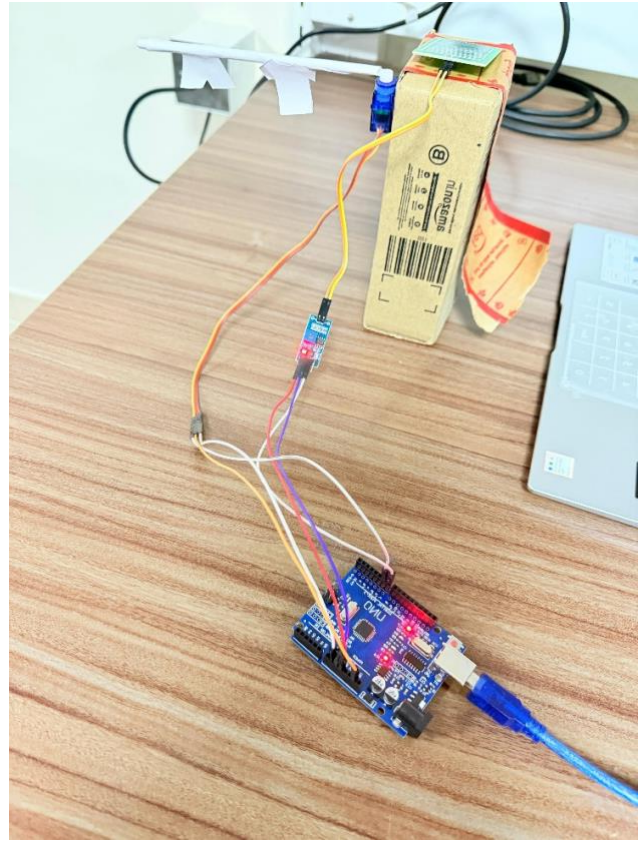
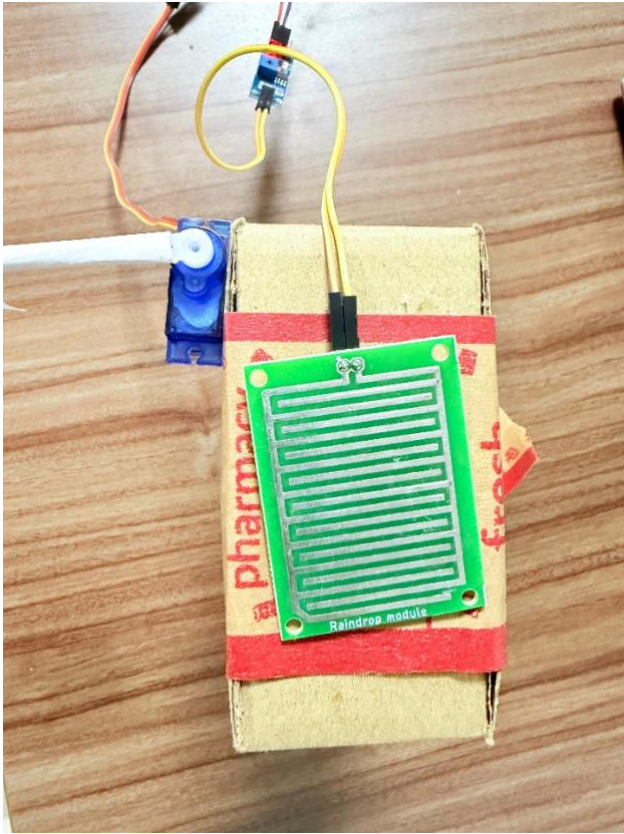
7.USB Cable



The power supply provides the necessary voltage to the Arduino and connected modules. It can be supplied through a computer USB port or an external 5V adapter. Stable power ensures reliable operation of the rain sensor and servo motor.

6.Images of Connection





7.Project Code

```
#include <Thread.h>

#include <Servo.h> // servo library

Servo s1;

int val = 0;

void setup()
{
    Serial.begin(9600);
    pinMode(7, INPUT); // Rain sensor output pin connected
    s1.attach(6);      // Servo Motor
}

void loop()
{
    val = digitalRead(7); // Rain sensor output pin connected
    Serial.println(val);  // see the value in serial monitor in Arduino
    IDE
    delay(100);

    if (val == 0)
    {
        s1.write(90); // Servo Rotate 90 Degree
        delay(500);   // delay time
    }

    if (val == 1)
```

```
{  
    s1.write(0);  // Servo Rotate 0 Degree  
}  
}
```

8.Output

Output Example:

- When **rain is detected** (sensor output = LOW), the servo motor rotates to **90°**, deploying the protective cover.
- When **no rain is detected** (sensor output = HIGH), the servo motor rotates back to **0°**, retracting the cover.
- The system successfully automates clothes protection without manual intervention.
- Serial Monitor displays real-time sensor values (0 for rain, 1 for no rain).

9.Applications

1. Household Use

In everyday households, clothes are often dried outdoors on balconies, terraces, or open areas. Sudden rainfall can make clothes wet again, forcing people to rush outside to collect them. This project solves that problem by automatically deploying a protective cover when rain is detected. It ensures convenience, saves time, and prevents repeated washing or drying.

2. Hostels and PG Accommodations

In shared living spaces, multiple residents dry clothes in common areas. It is impractical for each person to monitor their clothes continuously. A centralized rain sensor clothes protection system can safeguard everyone's clothes at once, reducing disputes and improving convenience in hostels or PGs.

3. Smart Homes

With the rise of IoT and smart home automation, this project can be integrated into a connected ecosystem. By adding Wi-Fi modules (ESP32/NodeMCU), the system can notify users via mobile apps when rain is detected. It can also be linked with other smart devices, such as automated windows or weather monitoring systems, making homes more intelligent and responsive.

4. Industrial and Commercial Use

Textile industries, laundries, and garment units often dry fabrics outdoors due to space constraints. Rain can damage materials and delay production schedules. Implementing this system ensures fabrics are automatically protected, reducing losses and improving efficiency. Small businesses like dry cleaners can also benefit by safeguarding customer clothes during drying.

5. Agricultural Applications

Farmers often sun-dry harvested crops, seeds, or grains in open fields. Rainfall can spoil these materials, leading to financial losses. A scaled-up version of this project can automatically cover crops when rain is detected, offering a low-cost protective solution for agriculture.

6. Public Utility Spaces

Community drying areas in apartment complexes or rural housing projects can benefit from this system. By installing a centralized rain sensor clothes protection mechanism, the entire community's drying space can be safeguarded, ensuring collective convenience and reducing the need for manual monitoring.

10. Conclusion

The rain sensor clothes protection project demonstrates how simple automation can solve everyday household problems. By combining a rain sensor, Arduino Uno, and servo motor, the system provides a reliable and cost-effective solution to protect clothes from unexpected rainfall. The project highlights the potential of embedded systems in smart home applications and can be further enhanced with IoT connectivity, predictive weather features, and sustainable energy sources.

11.Future Enhancements

- **IoT Integration:** Connect with Wi-Fi (ESP32/NodeMCU) to send mobile notifications when rain is detected.
- **Automatic Retraction System:** Use rails/pulleys to move clothes indoors instead of just deploying a cover.
- **Solar Power Support:** Add solar panels for sustainable energy supply.
- **Weather Prediction:** Integrate with online weather APIs to anticipate rainfall before it starts.
- **Multi-Sensor Setup:** Combine humidity and temperature sensors for smarter decision-making.

12.References

- EAZYTRONIC YouTube Channel – *Rain Sensor Clothes Protection Project* [https://youtu.be/ZBIK0Sj8uZo?si=P3vPVS_DWUqrPePo]
- Arduino Official Documentation – <https://www.arduino.cc>
- Servo Motor Datasheet (SG90).
- Rain Sensor Module Datasheet.